

ABSTRACT

This project presents the design and implementation of a Level 1 Signature-Based Antivirus Scanner developed using Python. The system scans files and directories recursively, computes cryptographic hashes (SHA256, MD5, SHA1), and compares them against a locally stored malware signature database. The scanner generates structured JSON reports and supports multi-threaded execution for improved performance. The system demonstrates fundamental antivirus working principles including file hashing, signature matching, and malware detection. This implementation is developed strictly for educational purposes and uses test malware signatures such as EICAR.

INTRODUCTION

This project presents a Python-based signature antivirus scanner that scans files and directories, computes cryptographic hashes (SHA256, MD5, SHA1), and compares them with a JSON-based malware signature database. The system generates structured reports and supports recursive directory scanning. It demonstrates the core principles of traditional antivirus detection mechanisms.

OBJECTIVES

- To understand the working principles of signature-based malware detection
- To implement cryptographic hashing for file fingerprinting
- To design a structured malware signature database
- To build a recursive file scanning engine
- To generate structured scan reports
- To simulate real-world antivirus architecture

DATABASE

The signature database is implemented using JSON format.

Features:

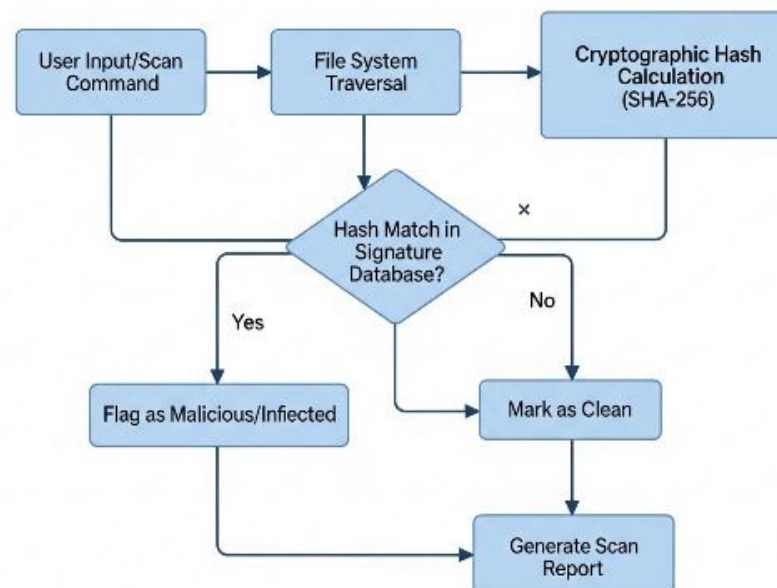
- Stores known malicious file hashes
- Supports SHA256, MD5, SHA1
- Lightweight and easily extendable

The scanner compares computed file hashes against stored malware signatures to detect infections.

METHODOLOGY

- Step 1: User Input
Command-line arguments specify the directory or file to scan.
- Step 2: Signature Loading
The JSON database is loaded into memory for quick lookup.
- Step 3: File Traversal
The scanner recursively walks through directories using OS file APIs.
- Step 4: Hash Calculation
Each file is read in chunks and hashed using:
 - I. SHA256
 - II. MD5
 - III. SHA1
- Step 5: Signature Matching
Computed hashes are compared with stored malware signatures.
- Step 6: Detection & Reporting
Match → Mark as Infected
No Match → Mark as Clean
Generate structured JSON report

Signature-Based Antivirus Scanner



CONCLUSION

The Level 1 Signature-Based Antivirus Scanner successfully demonstrates the core principles of traditional malware detection systems. The project illustrates how cryptographic hashing can be used to uniquely identify files and detect known malicious software.

Although signature-based detection is effective for known threats, it has limitations against new and unknown malware variants. This project provides a foundational understanding that can be extended to advanced detection mechanisms such as heuristic analysis, behavioral detection, and machine learning-based malware identification.

The system serves as a strong educational model for understanding antivirus architecture and cybersecurity fundamentals.

SOFTWARE SIMULATION

Technologies Used

- Python
- hashlib (cryptographic hashing)
- json (database handling)
- os (file system traversal)
- argparse (CLI interaction)
- YAML (configuration)

System Architecture

User Interface → Scanner Engine → Hasher Module → Signature Database → Reporter Module

Simulation Environment

- Developed and tested in controlled environment
- EICAR test signature used for validation
- Multi-threaded scanning implemented for performance improvement

REFERENCES

- EICAR Anti-Malware Test File Documentation
- Python hashlib Official Documentation
- OWASP Malware Detection Overview
- Cybersecurity Fundamentals – Signature-Based Detection Concepts
- OpenSSL Cryptographic Hash Functions Documentation